

HSP90 gives a scientific answer; rule value remains open

Can scientific reminders improve an autonomous agent's answer?

One exposed question. Four original answers.

A finite-time result, plus a concrete applicability problem.

Different data describe different aspects of molecular behavior

Structure references → geometry

Trajectory readouts → sampled changes over time

NMR measurements → model-dependent experimental evidence

We ask whether closed-seeded trajectories show opening

In apo human HSP90 α NTD, do closed-lid starts leave their initial conformation more readily than open-lid starts?

Which observed changes support that interpretation?

Henot et al. connect structure, NMR and finite trajectories

The ATP lid can adopt open-like and closed-like structures.

The paper reports stable open starts and heterogeneous closed starts.

A directional label is not the paper's full state definition.

Both arms received the same finite trajectory evidence

20 trajectories per starting group

20–1020 ns; 1,001 saved points per trajectory

Derived geometry, contacts, summaries and paper sources

Original coordinates and full native NOE inputs were absent.

The comparison changes reminders, not agent autonomy

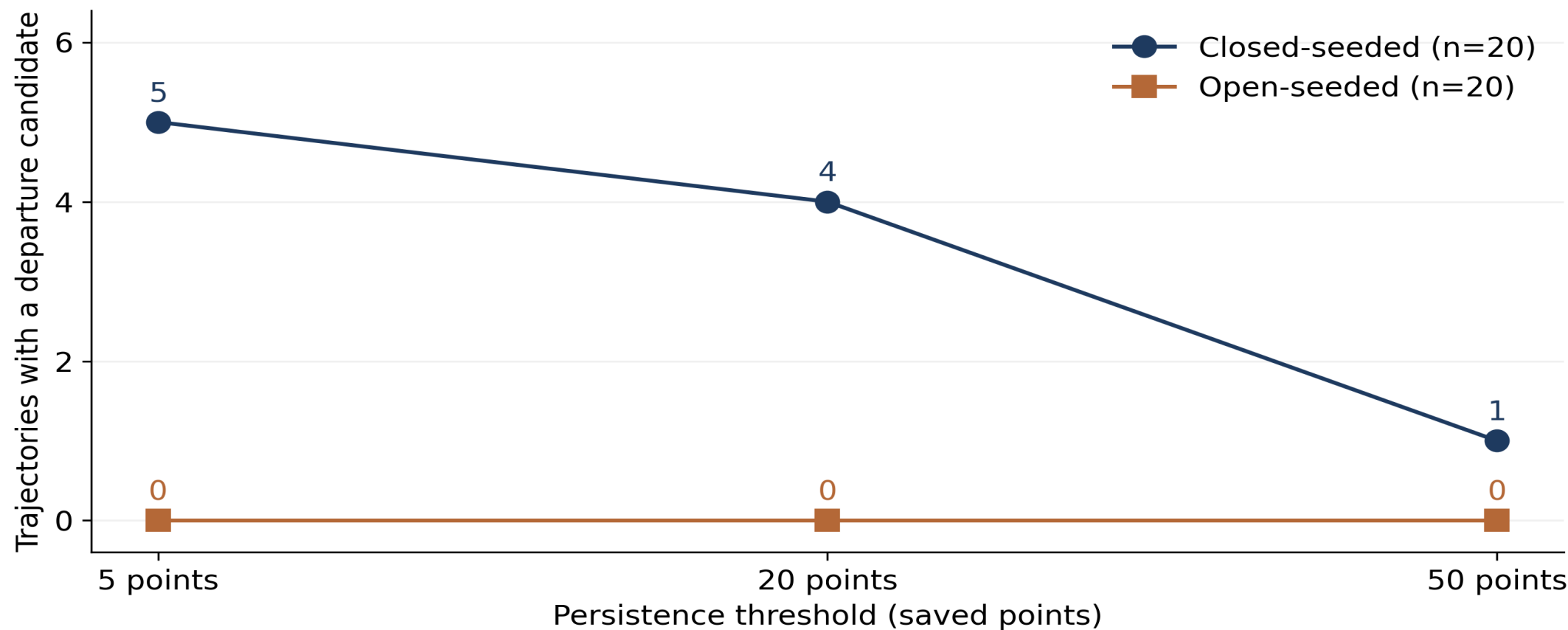
A: common data + tools + autonomous analysis

B: the same setup + selected rule text

Four fresh contexts: $A1 \rightarrow B1 \rightarrow B2 \rightarrow A2$

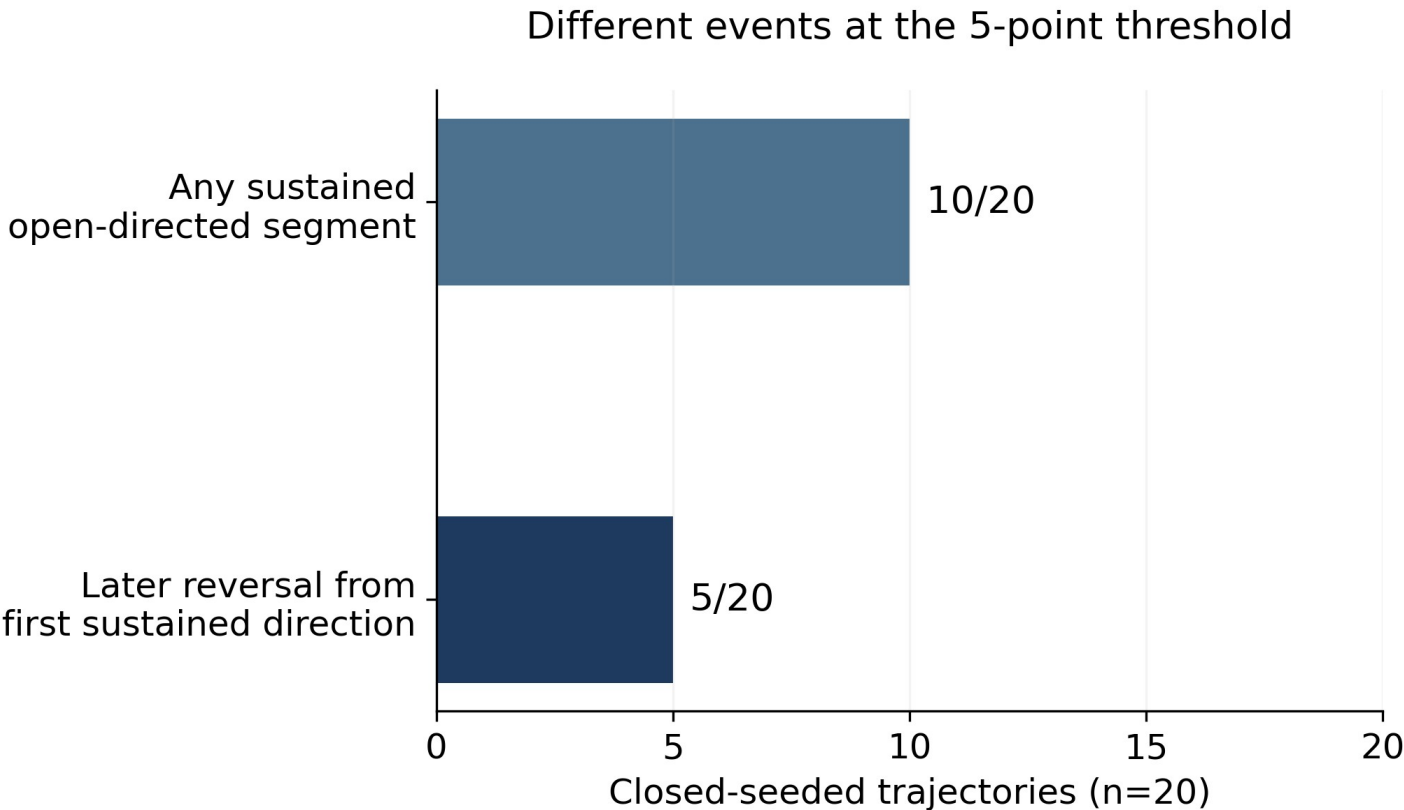
No live scientific hints; no deterministic enforcement arm.

Closed starts show more directional departure candidates



20-1020 ns; reference = first qualifying direction. Counts are not transition rates.

A sustained open segment is not the same as a transition



Narrow core labels are nearly empty

Closed-seeded: 1 C, 0 O

Open-seeded: 0 C, 2 O

20,020 saved points per group
All remaining points are U.

No trajectory contains both C and O.
This is a coverage limitation,
not proof of no physical transition.

Audited original outputs; events and saved-point labels must not be pooled into one population estimate.

Three answers recover the counts; one confuses thresholds

Answer	Observed behavior	Limitation
A1	Main counts + core coverage	Recovered from a failed request
B1	Main counts + core coverage	Unequal-bin wording; RMP terms
B2	Recomputed runs; added 10/20	No numeric core audit; RMP terms
A2	Conditions + some counts	Contradictory threshold counts

Four original outputs; total model cost \$0.0757458

Selected rules carried method conditions into the wrong context

Generic provenance and uncertainty checks carried RMP and dye-model requirements.

Both B answers imported RMP wording.

Missing rule IDs are an output-design issue, not a scientific error.

Our result is compatible with the paper, but not a full replication

Paper statement	This round supports
Open starts remain stable	No opposite sustained-direction candidates
Closed starts: about 7 / 9 / 4	Different event definitions; no numeric match claimed
Minor state from CPMG	Author-reported background, not a new MD estimate
Non-ergodic simulation	Finite-window, start-conditioned interpretation

Keep the result; resolve applicability before expanding

The scientific answer is auditable.

The incremental rule benefit is not established.

Current decision: stop automatic expansion.

Next science: compare directional segments with native NOE evidence.